

PETAR MLINARIĆ

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EXPERIENCE

University of Zagreb

Assistant professor

Mar 2025–present

Zagreb, Croatia

- Assistant professor at the Faculty of Science, Department of Mathematics.

Virginia Tech

Postdoctoral associate

Sep 2021–Dec 2024

Blacksburg, VA, United States

- Postdoctoral Associate in the Department of Mathematics at Virginia Tech.

Max Planck Institute for Dynamics of Complex Technical Systems

Research assistant, postdoctoral researcher

Sep 2014–Jul 2021

Magdeburg, Germany

- Started as a Ph.D. student at the Otto von Guericke University of Magdeburg and the graduate program “International Max Planck Research School (IMPRS) for Advanced Methods in Process and Systems Engineering (Magdeburg)”.
- Postdoctoral researcher since January 2020.

EDUCATION

Otto von Guericke University of Magdeburg, *Ph.D. in Applied Mathematics*

Thesis: Structure-Preserving Model Order Reduction for Network Systems

Supervisor: Prof. Dr. Peter Benner

Oct 2014–Jan 2020

University of Zagreb, *M.Sc. in Applied Mathematics*

Thesis: Approximation of Linear Dynamical Systems in $\mathcal{H}_2^{p \times m}(\mathbb{C}_+)$

Supervisor: Prof. Dr. Zlatko Drmač

Oct 2012–Jul 2014

University of Zagreb, *B.Sc. in Mathematics*

Oct 2009–Jul 2012

PREPRINTS

H. Kleikamp, P. Mlinarić, S. Rave, and F. Schindler, Data-Driven Model Order Reduction with pyMOR, *submitted*, July 2026

P. Mlinarić, S. Gugercin, and Z. Tomljanović, Optimal Damping for the 1D Wave Equation Using a Single Damper, *submitted*, September 2025

P. Mlinarić, P. Benner, and S. Gugercin, Interpolatory necessary optimality conditions for reduced-order modeling of parametric linear time-invariant systems, *accepted*, January 2024

JOURNAL PUBLICATIONS

G. Olikier, P. Mlinarić, P.-A. Absil, and A. Uschmajew, The tangent cone to the real determinantal variety: various expressions and four proofs, *Set-Valued and Variational Analysis*, March 2026

J.M. Maldonado, C. Drischler, R.J. Furnstahl, and P. Mlinarić, Greedy emulators for nuclear two-body scattering, *Physical Review C*, August 2025

P. Mlinarić, P. Benner, and S. Gugercin, Interpolatory $\mathcal{H}_2$ -optimality conditions for structured linear time-invariant systems, *SIAM Journal on Numerical Analysis*, April 2025

P. Mlinarić, C. A. Beattie, Z. Drmač, and S. Gugercin, IRKA is a Riemannian gradient descent method, *IEEE Transactions on Automatic Control (Early Access)*, November 2023

P. Mlinarić and S. Gugercin, A unifying framework for interpolatory $\mathcal{L}_2$ -optimal reduced-order modeling, *SIAM Journal on Numerical Analysis*, September 2023

P. Mlinarić and S. Gugercin, $\mathcal{L}_2$ -optimal reduced-order modeling using parameter-separable forms, *SIAM Journal on Scientific Computing*, April 2023

M. Hund, T. Mitchell, P. Mlinarić, and J. Saak, Optimization-based parametric model order reduction via $\mathcal{H}_2 \otimes \mathcal{L}_2$ first-order necessary conditions, *SIAM Journal on Scientific Computing*, June 2022

H.-J. Jongsma, P. Mlinarić, S. Grundel, P. Benner, and H. L. Trentelman, Model reduction of linear multi-agent systems by clustering with $\mathcal{H}_2$ and $\mathcal{H}_\infty$ error bounds, *Mathematics of Control, Signals, and Systems*, April 2018

BOOK CHAPTERS

P. Benner, S. Grundel, and P. Mlinarić, Clustering-based model order reduction for nonlinear network systems, In: P. Benner, T. Breiten, H. Faßbender, M. Hinze, T. Stykel, R. Zimmermann (eds), *Model Reduction of Complex Dynamical Systems*, August 2021

P. Mlinarić, S. Rave, and J. Saak, Parametric model order reduction using pyMOR, In: P. Benner, T. Breiten, H. Faßbender, M. Hinze, T. Stykel, R. Zimmermann (eds), *Model Reduction of Complex Dynamical Systems*, August 2021

CONFERENCE PROCEEDINGS PUBLICATIONS

P. Mlinarić, Z. Sokolij, G. Talla, U. Meyer-Baese, A. Stadlbauer, H. Malberg, C.-H. Na, K. Juetten, B. Wiestler, and A. Meyer-Baese, Clustering-based model reduction for glioma graph networks, In: B.S. Gimi, A. Krol (eds), *Medical Imaging 2025: Clinical and Biomedical Imaging*, April 2025

L. Balicki, P. Mlinarić, S. Rave, and J. Saak, System-theoretic model order reduction with pyMOR, *Proceedings in Applied Mathematics and Mechanics*, November 2019

M. Hund, P. Mlinarić, and J. Saak, An $\mathcal{H}_2 \otimes \mathcal{L}_2$ -optimal model order reduction approach for parametric linear time-invariant systems, *Proceedings in Applied Mathematics and Mechanics*, December 2018

P. Mlinarić, T. Ishizaki, A. Chakraborty, S. Grundel, P. Benner, and J. Imura, Synchronization and aggregation of nonlinear power systems with consideration of bus network structures, *Proceedings of the European Control Conference (ECC)*, June 2018

P. Benner, S. Grundel, and P. Mlinarić, Stability preserving model reduction for linearly coupled linear time-invariant systems, *Proceedings in Applied Mathematics and Mechanics*, October 2016

P. Mlinarić, S. Grundel, and P. Benner, Clustering-based model order reduction for multi-agent systems with general linear time-invariant Agents, *Proceedings of the 22nd International Symposium on Mathematical Theory of Networks and Systems (MTNS)*, July 2016

P. Mlinarić, S. Grundel, and P. Benner, Efficient model order reduction for multi-agent systems using QR decomposition-based clustering, *Proceedings of the 54th IEEE Conference on Decision and Control*, December 2015

PROFESSIONAL SERVICE

pyMOR School and User Meeting 2024 co-organizer 26–30 Aug 2024
The sixth pyMOR School, taking place at U Münster (<https://2024.school.pymor.org/>).

pyMOR School 2023 co-organizer 18–22 Sep 2023
The fifth pyMOR School, taking place at TU Berlin (<https://2023.school.pymor.org/>).

SIAM Southeastern Atlantic Section minisymposium co-organizer 25–26 Mar 2023
A minisymposium at the SIAM SEAS23 conference in Blacksburg, US.

SIAM CSE23 pyMOR minitutorial co-organizer 1 Mar 2023
A pyMOR minitutorial at the SIAM CSE23 conference in Amsterdam, The Netherlands.

SIAM CSE23 minisymposium co-organizer 26 Feb–3 Mar 2023
A minisymposium at the SIAM CSE23 conference in Amsterdam, The Netherlands.

pyMOR School 2022 co-organizer 22–26 Aug 2022
The fourth pyMOR School, taking place at MPI Magdeburg (<https://2022.school.pymor.org/>).

Virginia Tech Department of Mathematics seminar organizer Aug 2022–Apr 2024
Organized the Applied Numerical Analysis seminar at the Department of Mathematics at Virginia Tech.

Virginia Tech Postdoctoral Association board member May 2022–Jul 2024
Held positions as secretary (2022/23) and communications officer (2023/24).

pyMOR School 2021 co-organizer 4–8 Oct 2021
The third pyMOR School, taking place at WWU Münster (<https://2021.school.pymor.org/>).

pyMOR Online Course 2020 co-organizer 28 Sep–2 Oct 2020
The second pyMOR School in an online format (<https://2020.school.pymor.org/>).

SAMM 2020 co-organizer 27–31 Jul 2020
The 7th GAMM Juniors’ Summer School on Applied Mathematics and Mechanics (SAMM) with the topic “Learning Models from Data: Model Reduction, System Identification and Machine Learning” (<https://www.mpi-magdeburg.mpg.de/csc/events/samm20>).

pyMOR School 2019 co-organizer 7–11 Oct 2019
The first pyMOR School, taking place at MPI Magdeburg (<https://2019.school.pymor.org/>).

Held positions as president (2015/16) and secretary (2016/17).

Reviewer

Advances in Computational Mathematics ($\times 2$), Applied Mathematics Modeling, Applied Mathematics and Computation, Automatica, Computer Methods in Applied Mechanics and Engineering, European Control Conference, European Journal of Control, GAMM Archive for Students, IEEE Conference on Decision and Control, IEEE Control Systems Letters, IEEE Transactions on Automatic Control ($\times 5$), International Symposium on Mathematical Theory of Networks and Systems ($\times 2$), Journal of Applied Mathematics and Mechanics, Journal of Computational Physics, Journal of Computational and Applied Mathematics, MDPI Mathematics ($\times 3$), Mathematical Communications, Research in the Mathematical Sciences, SIAM Journal on Control and Optimization, SIAM Journal on Mathematics of Data Science, SIAM Journal on Matrix Analysis and Applications, SIAM Journal on Numerical Analysis, SIAM Journal on Scientific Computing ($\times 3$), SoftwareX, Systems & Control Letters ($\times 4$), Transactions on Mathematical Software.

TEACHING EXPERIENCE

Otto von Guericke University of Magdeburg

Model Order Reduction of Dynamical Systems (Summer 2017, exercises)

Virginia Tech

Math 2114: Introduction to Linear Algebra (Spring 2022, Spring 2023)

Math 4445: Introduction to Numerical Analysis (Spring 2024)

RESEARCH VISITS AND TRAININGS

UCLouvain

2–6 Feb 2026

Visited Prof. Dr. Pierre-Antoine Absil for one week within an ERASMUS+ project.

Josip Juraj Strossmayer University of Osijek

12–16 Sep 2022

Visited Prof. Dr. Zoran Tomljanović for one week.

Josip Juraj Strossmayer University of Osijek

17–23 Dec 2021

Visited Prof. Dr. Zoran Tomljanović for one week within an ERASMUS+ project.

ICERM semester program

27 Jan–1 May 2020

Participated in the semester program “Model and dimension reduction in uncertain and dynamic systems” (<https://icerm.brown.edu/programs/sp-s20/>).

Groningen Autumn School on Model Order Reduction

30 Oct–2 Nov 2017

Participated in the fall school in Groningen, The Netherlands.

Tokyo Institute of Technology

7 Jul–15 Sep 2017

Visited Prof. Dr. Jun-ichi Imura’s research group for two months.

University of Groningen

20–21 Jan 2016

Visited Prof. Dr. Harry Trentelman for two days.

The Third International School on Model Reduction for Dynamical Control Systems

5–9 Oct 2015

Participated in the summer school in Dubrovnik, Croatia.

7th Elgersburg School

1–7 Mar 2015

Participated in the winter school in Elgersburg, Germany.

DAAD International School on Linear Optimal Control of Dynamic Systems

23–28 Sep 2013

Participated in the summer school in Osijek, Croatia.

AWARDS

GAMM Junior

2017–2020

Three-year position awarded yearly to ten young academics working in the field of applied mathematics and mechanics (<https://www.gamm-juniors.de/>).

INVITED MINISYMPOSIUM TALKS

27th Conference of the International Linear Algebra Society (ILAS 2026),

18–22 May 2026

Blacksburg, VA, USA

Least-squares Rational Approximation Using Riemannian Optimization

26th International Symposium on Mathematical Theory of Networks and Systems (MTNS 2024), Cambridge, UK

19–23 Aug 2024

TF-IRKA is also a Riemannian Gradient Descent Method

European Conference on Numerical Mathematics and Advanced Applications (ENUMATH 2023) , Lisbon, Portugal <i>Parametric Reduced-order Modeling via Nonlinear Least Squares</i>	4–8 Sep 2023
25th Conference of the International Linear Algebra Society (ILAS 2023) , Madrid, Spain <i>Optimal Reduced-order Modeling for Structured Linear Systems</i>	12–16 Jun 2023
SIAM Conference on Computational Science and Engineering (CSE23) , Amsterdam, The Netherlands <i>Least-squares Parametric Reduced-order Modeling</i>	26 Feb–3 Mar 2023
5th Annual Meeting of the SIAM Texas-Louisiana Section (SIAM TXLA22) , Houston, TX, USA <i>Least-squares Parametric Reduced-order Modeling</i>	4–6 Nov 2022
SIAM Conference on Control and Its Applications (CT21) , virtual <i>Clustering-Based Model Order Reduction for Water Networks</i>	19–21 Jul 2021
European Control Conference (ECC18) , Limassol, Cyprus <i>Synchronization and Aggregation of Nonlinear Power Systems with Consideration of Bus Network Structures</i>	12–15 Jun 2018
22nd International Symposium on Mathematical Theory of Networks and Systems (MTNS 2016) , Minneapolis, MN, USA <i>Clustering-Based Model Order Reduction for Multi-Agent Systems with General Linear Time-Invariant Agents</i>	12–15 Jul 2016

CONFERENCE AND WORKSHOP TALKS

Challenges, Opportunities, and New Horizons in Rational Approximation , Banff, Canada <i>Riemannian Optimization over Rational Functions</i>	6–11 Apr 2025
4th Workshop on Optimal Control of Dynamical Systems and Applications , Villány, Hungary <i>Optimal Damping for the Wave Equation</i>	26–28 Feb 2025
Model Reduction and Surrogate Modeling (MORe2024) , San Diego, CA, USA <i>Riemannian Optimization over the Manifold of Rational Functions</i>	9–13 Sep 2024
Nonlinear Model Reduction for Control , Blacksburg, VA, USA <i>$\mathcal{H}_2 \otimes \mathcal{L}_2$-optimal Model Order Reduction for Quadratic-bilinear Systems</i>	22–26 May 2023
SIAM Southeastern Atlantic Section Annual Meeting , Blacksburg, VA, USA <i>$\mathcal{L}_2$-optimal Reduced-order Modeling of Parametric Stationary Problems</i>	25–26 Mar 2023
Model Reduction and Surrogate Modeling (MORE) , Berlin, Germany <i>Structured $\mathcal{L}_2$-Optimal Parametric Model Order Reduction</i>	19–23 Sep 2022
11th Conference on Applied Mathematics and Scientific Computing (ApplMath22) , Brijuni, Croatia <i>Iterative Rational Krylov Algorithm and Riemannian Optimization</i>	5–9 Sep 2022
New Trends in Computational Science in Engineering and Industrial Mathematics , Magdeburg, Germany <i>Is IRKA a Riemannian Optimization Method?</i>	1–2 Jul 2022
Spring 2020 Reunion Event , Providence, RI, USA <i>$\mathcal{L}_2$-Optimal Reduced-Order Modeling</i>	23 May–10 Jun 2022
SIAM Conference on Uncertainty Quantification (UQ22) , Atlanta, GA, USA <i>$\mathcal{L}_2$-Optimal Model Order Reduction for Parametric Stationary Problems</i>	12–15 Apr 2022
3rd Workshop on Optimal Control of Dynamical Systems and Applications , Osijek, Croatia <i>Structured $\mathcal{L}_2$-Optimal Parametric Model Order Reduction</i>	28 Mar–1 Apr 2022
Tenth Conference on Applied Mathematics and Scientific Computing (ApplMath20) , Brijuni, Croatia <i>Optimization-Based Parametric Model Order Reduction via Wilson-Type $\mathcal{H}_2 \otimes \mathcal{L}_2$ Conditions</i>	14–18 Sep 2020
4th Workshop on Model Reduction of Complex Dynamical Systems (MODRED 2019) , Graz, Austria <i>Clustering-Based Model Order Reduction for Nonlinear Network Systems</i>	28–30 Aug 2019
90th GAMM Annual Meeting , Vienna, Austria <i>$\mathcal{H}_2$-Optimal Model Order Reduction of Interconnected Systems</i>	18–22 Feb 2019

International Workshop on Optimal Control of Dynamical Systems and Applications , Osijek, Croatia <i>H₂-Optimal Model Order Reduction for Port-Hamiltonian Systems</i>	20–22 Jun 2018
Model Reduction of Parameterized Systems IV (MoRePaS 2018) , Nantes, France <i>H₂-Optimal Structure-Preserving Model Order Reduction for Second-Order Systems</i>	10–13 Apr 2018
89th GAMM Annual Meeting , Munich, Germany <i>Clustering-Based Model Order Reduction for Nonlinear Multi-Agent Systems</i>	19–23 Mar 2018
Reduced Basis (and Friends) Summer School 2017 , Goslar, Germany <i>H₂-Optimal Model Order Reduction and Reduced Basis Methods</i>	19–22 Sep 2017
88th GAMM Annual Meeting , Weimar, Germany <i>H₂-Optimal Model Order Reduction for Second Order Systems</i>	6–10 Mar 2017
3rd Workshop on Model Reduction of Complex Dynamical Systems (MODRED 2017) , Odense, Denmark <i>Model Reduction of Network Systems by Clustering</i>	11–13 Jan 2017
Reduced Basis Summer School 2016 , Hedersleben, Germany <i>Model Order Reduction for Multi-Agent Systems</i>	4–7 Oct 2016
Workshop on Model Reduction Methods and Optimization , Opatija, Croatia <i>Almost Almost-Equitableness and Clustering-Based Error Bounds</i>	20–21 Sep 2016
6th Croatian Mathematical Congress , Zagreb, Croatia <i>Clustering-Based Model Order Reduction for Multi-Agent Systems</i>	14–17 Jun 2016
Joint Annual Meeting of DMV and GAMM , Braunschweig, Germany <i>Stability Preserving Model Reduction for Linearly Coupled Linear Time-Invariant Systems</i>	7–11 Mar 2016
10th Elgersburg Workshop , Elgersburg, Germany <i>Efficient Model Order Reduction for Multi-Agent Systems Using Graph Partitions</i>	7–11 Feb 2016
54th IEEE Conference on Decision and Control (CDC) , Osaka, Japan <i>Efficient Model Order Reduction for Multi-Agent Systems Using QR Decomposition-Based Clustering</i>	15–18 Dec 2015
Reduced Basis Summer School 2015 , Hegne, Germany <i>Efficient Model Order Reduction for Multi-Agent Systems Using Graph Partitions</i>	14–18 Sep 2015

TALKS AT INSTITUTIONS

Department of Mathematical Engineering (INMA) Seminar , UCLouvain, Louvain-la-Neuve, Belgium <i>IRKA Is a Riemannian Gradient Descent Method</i>	3 Feb 2026
Scientific Colloquium , University of Zagreb, Zagreb, Croatia <i>The Unintended Legacy of G. H. Hardy</i>	14 Jan 2026
Numerical Mathematics and Scientific Computing Seminar , University of Zagreb, Zagreb, Croatia <i>H₂-optimalna redukcija reda modela i Riemannova optimizacija</i>	15 May 2025
Applied Numerical Analysis Seminar , Virginia Tech, Blacksburg, VA, USA <i>What is the Optimal Damper Position for the Wave Equation?</i>	13 Oct 2023
Applied Numerical Analysis Seminar , Virginia Tech, Blacksburg, VA, USA <i>Optimal Reduced-order Modeling of Structured Linear Systems</i>	24 Mar 2023
Applied Numerical Analysis Seminar , Virginia Tech, Blacksburg, VA, USA <i>Is IRKA a Riemannian Optimization Method?</i>	14 Oct 2022
Optimization and Applications Seminar , Josip Juraj Strossmayer University of Osijek, Osijek, Croatia <i>Interpolatory Conditions for $\mathcal{L}_2$-optimal Reduced-order Modeling</i>	14 Sep 2022
Applied Numerical Analysis Seminar , Virginia Tech, Blacksburg, VA, USA <i>Structured $\mathcal{L}_2$-optimal Parametric Model Order Reduction</i>	1 Apr 2022
Applied Numerical Analysis Seminar , Virginia Tech, Blacksburg, VA, USA <i>H₂ $\otimes$ $\mathcal{L}_2$-optimal Parametric Model Order Reduction via Wilson-type conditions</i>	24 Sep 2021
Optimization and Applications Seminar , Josip Juraj Strossmayer University of Osijek, Osijek, Croatia <i>Model Order Reduction for Parametric Stationary Problems</i>	22 Dec 2021

Optimization and Applications Seminar, Josip Juraj Strossmayer University of Osijek, Osijek, Croatia <i>System-theoretic Model Order Reduction in pyMOR</i>	2 May 2019
Numerical Mathematics and Scientific Computing Seminar, University of Zagreb, Zagreb, Croatia <i>Optimalna $\mathcal{H}_2$ redukcija s očuvanjem strukture za linearne sustave</i>	24 Apr 2019
Imura Lab Seminar, Tokyo Institute of Technology, Tokyo, Japan <i>$\mathcal{H}_2$-optimal Model Order Reduction of Network Systems</i>	21 Jul 2017
Computational Methods in Systems and Control Seminar, Max Planck Institute for Dynamics of Complex Technical Systems, Magdeburg, Germany <i>$\mathcal{H}_2$-optimal Model Order Reduction of Structured Linear Systems</i>	11 Jul 2017
Seminar Regelungssysteme, Technische Universität Berlin, Berlin, Germany <i>Efficient Structure-preserving Model Order Reduction for Multi-agent Systems</i>	22 Sep 2015
Oberseminar Angewandte Mathematik, Westfälische Wilhelms-Universität Münster, Münster, Germany <i>Model Order Reduction of Input-Output Systems</i>	17 Jun 2015